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Task 5: Voice-Activated Personal Assistant Using Python

1. Introduction

A voice-activated personal assistant is a software application that can understand voice commands and respond using speech output. Such systems use speech recognition and text-to-speech techniques to interact with users. In this task, a simple voice-activated personal assistant is developed using Python to perform basic operations.

2. Objective

The objectives of this task are:

- To develop a voice-activated personal assistant using Python
 - To implement speech recognition for taking user commands
 - To generate voice responses using text-to-speech
 - To perform basic tasks such as greeting the user and telling the current time
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3. Tools and Technologies Used

- **Programming Language:** Python
 - **Libraries Used:**
 - SpeechRecognition – for converting speech to text
 - pyttsx3 – for text-to-speech conversion
 - datetime – for fetching current date and time
 - **IDE:** Visual Studio Code
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4. System Description

The voice assistant listens to user commands through a microphone and converts the speech into text. Based on the recognized command, the assistant performs predefined tasks and responds using audio output. The system runs continuously until the user gives an exit command.

5. Methodology

1. The speech recognition module was initialized to capture voice input from the microphone.
 2. The text-to-speech engine was configured to provide audio responses.
 3. The assistant listened for voice commands using a microphone.
 4. Recognized speech was converted into text and processed using conditional statements.
 5. Based on the command, appropriate actions were performed such as greeting the user, telling the time, or exiting the program.
 6. The assistant responded to the user using voice output.
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6. Results and Observations

- The assistant successfully recognized voice commands using a wired earphone microphone.
 - It responded correctly to commands such as greeting, asking the time, and requesting its name.
 - The system produced clear voice output using text-to-speech.
 - In situations where a microphone was unavailable, keyboard input was used as an alternative for demonstration purposes.
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7. Conclusion

This task successfully demonstrated the development of a voice-activated personal assistant using Python. It helped in understanding speech recognition, text-to-speech conversion, and basic human-computer interaction. The project enhanced practical knowledge of implementing voice-based applications in Python.

8. Output

- Python program (voice_assistant.py)
- Terminal output showing recognized commands
- Audio responses generated by the assistant